

Challenge 3: Caccetta–Häggkvist Conjecture

Every simple digraph in which each vertex has an out-neighbour contains a directed cycle, but not necessarily a short one. The Caccetta–Häggkvist conjecture predicts exactly how the minimum out-degree forces short cycles. The natural example is the digraph on n cyclically ordered vertices in which every vertex is joined to the next r vertices: it has minimum out-degree r , and its shortest directed cycle has length $\lceil n/r \rceil$. The conjecture asserts that no digraph with minimum out-degree r does better.

Conjecture 1.1 (Caccetta–Häggkvist Conjecture [1]). Let $r \in \mathbb{N}$ with $r \geq 1$. Every simple directed graph on n vertices with minimum out-degree at least r has a cycle of length at most $\lceil n/r \rceil$.

For $r = 1$ the statement is elementary. The conjecture was proved for $r = 2$ by Caccetta and Häggkvist [1], for $r = 3$ by Hamidoune [2], and for $r = 4$ and $r = 5$ by Hoàng and Reed [3]. Shen proved it for every r under the additional assumption $n \geq 2r^2 - 3r + 1$ [4].

References

- [1] L. Caccetta and R. Häggkvist. “On minimal digraphs with given girth”. In: *Proceedings of the Ninth Southeastern Conference on Combinatorics, Graph Theory, and Computing (Boca Raton, 1978)*. Vol. 21. Congressus Numerantium. Utilitas Mathematica, 1978, pp. 181–187.
- [2] Y. O. Hamidoune. “A note on minimal directed graphs with given girth”. In: *Journal of Combinatorial Theory, Series B* 43.3 (1987), pp. 343–348.
- [3] C. T. Hoàng and B. Reed. “A note on short cycles in digraphs”. In: *Discrete Mathematics* 66.1–2 (1987), pp. 103–107.
- [4] J. Shen. “On the girth of digraphs”. In: *Discrete Mathematics* 211.1–3 (2000), pp. 167–181.